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Algorithm 4 Meta-chain Distillation

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1: set  $S_o = \emptyset$ ,  $\mathbf{Z}^{(0)} = \mathbf{0}$ ,  $\iota(x) = 0$  for all  $x \in S$ ,  
2:  $T_{\text{dist}} = O(M^2(\log K)^2\varepsilon^{-2})$ ,  $T_{\text{thres}} = \tilde{O}(M\varepsilon^{-1})$   
3:  $\eta = \Theta(K)$ ,  $\beta = \Theta(\log(M/\varepsilon))$   
4: while  $\iota^{-1}(0) \neq \emptyset$  do (cluster labeling)  
5:   draw  $X_0 \in \iota^{-1}(0)$   
6:    $S_o \leftarrow S_o \cup \{X_0\}$ ,  $\iota(X_0) \leftarrow X_0$   
7:   for  $t = 1, \dots, T_0$  do  
8:     generate  $X_t^\varepsilon \sim p^\varepsilon(\cdot | X_{t-1}^\varepsilon)$   
9:      $\iota(X_t^\varepsilon) \leftarrow X_0$   
10:  end for  
11: end while  
12: for  $t = 1, 2, \dots$  do (data collection)  
13:   if  $X_t^\varepsilon \in S_o$  then  
14:      $Y_0^{(t)}, Y_1^{(t_{\text{prev}})} \leftarrow X_t^\varepsilon$   
15:      $t_{\text{prev}} \leftarrow t$   
16:   else  
17:      $Y_0^{(t)}, Y_1^{(t)} \leftarrow \iota(X_t^\varepsilon)$   
18:   end if  
19: end for  
20: for  $t = 1, \dots, T_{\text{dist}}$  do (distillation)  
21:    $\mathbf{Z}^{(t)} = \mathbf{Z}^{(t-1)} - \eta \nabla \mathbb{E}_{Y_0, Y_1} [-\log \hat{p}_{\mathbf{Z}^{(t-1)}}(Y_1 | Y_0)]$   
22:   if  $t = T_{\text{thres}}$  then  
23:      $z_{k\ell}^{(T_{\text{thres}})} \leftarrow -\infty$  if  $\hat{q}_{k\ell}^{(T_{\text{thres}})} < c_{\text{thres}}\varepsilon/M$   
24:   end if  
25: end for  
26:  $z_{k\ell}^+ \leftarrow z_{k\ell}^{(T_{\text{dist}})} + \beta$  for  $\ell \neq k$  (time rescaling)  
27: return  $\mathbf{Z}^+$ 
```

A Additional Related Works

Theoretical Analysis of CoT. Some theoretical works have focused on the expressivity of CoT (Feng et al., 2023a; Merrill and Sabharwal, 2023; Chiang et al., 2023; Li et al., 2024b), analysis of optimization and estimation ability (Li et al., 2024a; Hu et al., 2024; Kim and Suzuki, 2024), or in-context learning ability (Li et al., 2023; Bhattamishra et al., 2024). More closely related to our paper, Sanford et al. (2024b,a); Abbe et al. (2024) study the algorithmic reasoning capabilities of CoT or scratchpad transformers for certain computational or graph-based tasks. Also, Nichani et al. (2024) analyze how simple transformer models learn latent causal structure within the data.